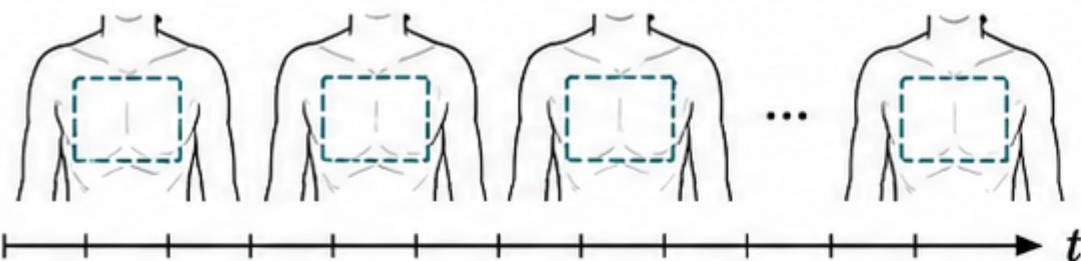


camera-derived respiratory evidence

RGB frames + chest ROI

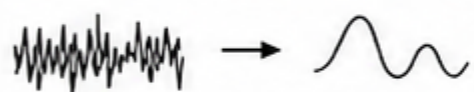


motion operator cues
(fixed observation classes)

optical-flow cues (OF / OF-bridge)

OF

OF-bridge



frame-difference cues (DoF / DoF-bridge)

DoF

DoF-bridge



profile-shift cues (PID variants)

PID-linear

PID-quadratic

PID-cubic



PID-consensus

candidate evidence pool
(heterogeneous views)

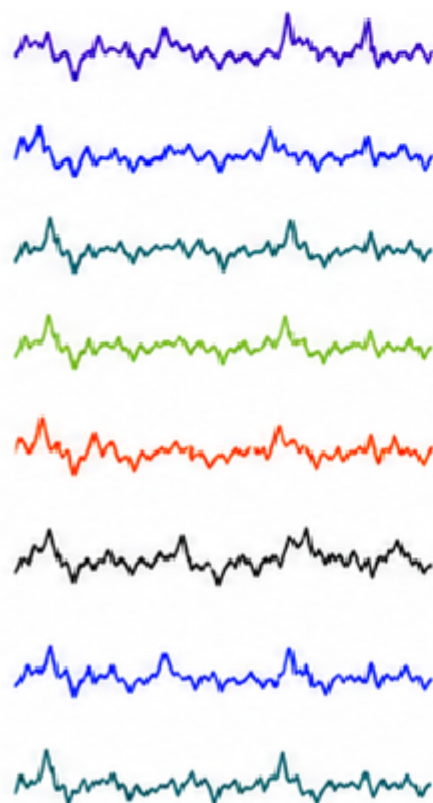
view-level evidence summaries

agreement

alias risk

nuisance

observability



role-specific observation
weights W_t

evidence $\rightarrow$
state-role responsibilities

$h^{(1)}$ $h^{(2)}$ b r
(osc) (harm) (base) (res)



adaptive observation law (label-free)

π_t

trust

$R_{eff,t}$

effective
noise

W_t

role
weights

Q_{osc}, Q_{aper}

osc/residual
process scales

abstention
pressure

PARH-OSSM latent respiratory state decomposition

state-space inference

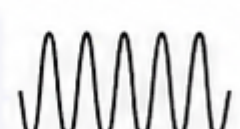
KF/RTS smoothing + Student-t innovation weights

$h^{(1)}$
(osc)



oscillator drive
(dominant rhythm)

$h^{(2)}$
(harm)



harmonic asymmetry
(second harmonic)

b
(base)



baseline /
projection drift
(slow component)

r
(res)



aperiodic residual
(respiratory residual)

z_{osc}

+

z_{morph}

+

decoupled readouts
(role-aligned outputs)

diagnostics (side output)

NIS /
innovation

confidence /
calibration

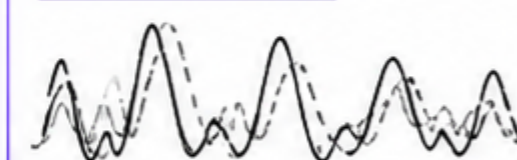
observability
flags

failure
flags

state / readout consistency

Readout 1: oscillatory (rate & timing)

$z_{osc} = h_t^{(1)} + h_t^{(2)}$ rate / BPM



Readout 2: morphology (waveform)

$z_{full} = z_{osc} + z_{morph}$ waveform

